

Intelligent Systems Development

REPORT

Comparison and Improved Models

Cross-dataset comparison of the three implementations, and the M1–M4
architecture-evolution experiment

Assignment 4 — Comparing CNN Implementations

September 2026

Contents

Part 1 — Three implementations, three datasets	2
1.1 Every run, in one table	2
1.2 Parameter counts agree with the hand calculation	8
1.3 The component table (lecture slide 28)	8
1.4 Function-to-API mapping (lecture slide 18)	9
1.5 The eight questions (lecture slide 24)	10
Part 2 — Improved CNN models (tutorial §45)	10
2.1 The four models	10
2.2 Training all four	12
2.3 Results	14
2.4 Problem and solution	19
Conclusions	22
On the three implementations	22
On matching the model to the data	22
On improved CNN models	22
Honest limits of these experiments	23

This notebook closes the assignment. It has two halves:

Part 1 — the three-implementation comparison, gathered across all three datasets: the results tables, the component table from lecture slide 28, the visibility table from slide 23, and the answers to the eight questions on slide 24.

Part 2 — improved CNN models, the experiment tutorial §45 asks for:

$$M_1 : \text{Conv} + \text{ReLU} + \text{Pool} \quad M_2 : M_1 + \text{BatchNorm} \quad M_3 : M_2 + \text{Residual} \quad M_4 : M_3 + \text{Attention}$$

Each model is the previous one plus exactly one mechanism. The tutorial's framing is that architectural history is not a list of names to memorise but a sequence of responses to specific limitations:

new architecture = old architecture + a mechanism addressing a limitation

The point, as §45 says explicitly, is *not* to obtain the highest accuracy. It is to make the evolution measurable — accuracy, parameters, training time and representation behaviour together.

```
import os, sys, json, glob

os.environ.setdefault("TF_CPP_MIN_LOG_LEVEL", "2")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

sys.path.insert(0, os.path.abspath("."))
import ass4_utils as U
import variants as V          # the M1..M4 models, shared with prefill_variants.py

U.set_seed(U.SEED)

import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import TensorDataset, DataLoader

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")
plt.rcParams["figure.dpi"] = 110
print("device:", DEVICE, "|", torch.cuda.get_device_name(0) if torch.cuda.is_available() else "CPU
↪ only")
```

| device: cuda | NVIDIA GeForce RTX 3060 Laptop GPU

Part 1 — Three implementations, three datasets

1.1 Every run, in one table

Loaded from the JSON each of the first three notebooks wrote. **Run notebooks 01–03 before this one.**

```
files = ["results/01_diabetes130.json", "results/02_mnist.json", "results/03_cifar10.json"]
missing = [f for f in files if not os.path.exists(f)]
if missing:
    raise FileNotFoundError(
        f"missing {missing} — run notebooks 01-03 first, they write these files."
    )

payloads, rows = {}, []
for f in files:
```

```
d = json.load(open(f))
payloads[f] = d
for key in ("results", "full_results"):
    for r in d.get(key, []):
        rows.append({k: v for k, v in r.items() if k not in ("history", "confusion")})

all_runs = pd.DataFrame(rows)
display(all_runs.round(4))
```

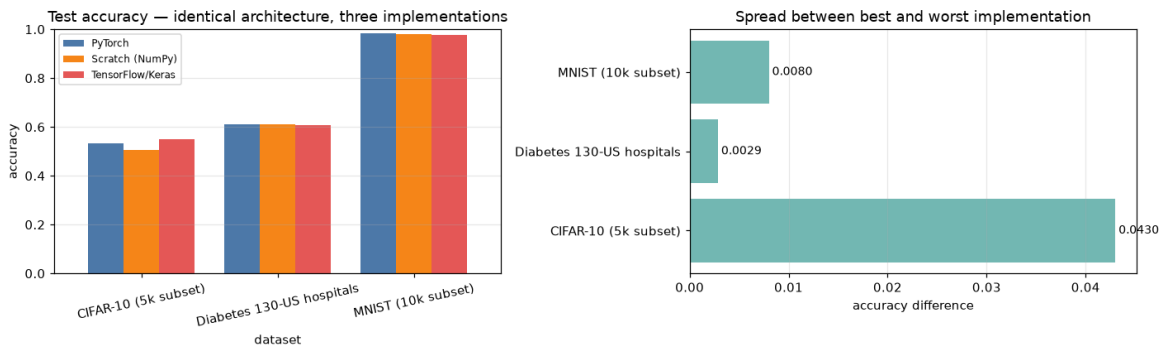
	framework	dataset	model	n_params	epochs	train_seconds	train_loss	test_accuracy	precision_macro	recall_macro	f1_macro
0	Scratch (NumPy)	Diabetes 130-US hospitals	MLP 179-128-64-3	31491	15	6.04	0.8237	0.6094	0.4798	0.3728	0.3450
1	TensorFlow/Keras	Diabetes 130-US hospitals	MLP 179-128-64-3	31491	15	7.78	0.8174	0.6077	0.4839	0.3842	0.3686
2	PyTorch	Diabetes 130-US hospitals	MLP 179-128-64-3	31491	15	9.17	0.8135	0.6105	0.5129	0.3860	0.3694
3	Scratch (NumPy)	MNIST (10k subset)	CNN 2conv+fc	20490	5	36.24	0.0525	0.9810	0.9813	0.9810	0.9810
4	TensorFlow/Keras	MNIST (10k subset)	CNN 2conv+fc	20490	5	4.09	0.0665	0.9755	0.9765	0.9754	0.9756
5	PyTorch	MNIST (10k subset)	CNN 2conv+fc	20490	5	1.95	0.0569	0.9835	0.9838	0.9835	0.9836
6	PyTorch	MNIST (full 60k)	CNN 2conv+fc	20490	5	9.06	0.0299	0.9875	0.9875	0.9875	0.9874
7	TensorFlow/Keras	MNIST (full 60k)	CNN 2conv+fc	20490	5	19.19	0.0275	0.9855	0.9855	0.9854	0.9854
8	Scratch (NumPy)	CIFAR-10 (5k subset)	SmallCNN	545098	5	65.23	1.0515	0.5055	0.5216	0.5080	0.5032
9	TensorFlow/Keras	CIFAR-10 (5k subset)	SmallCNN	545098	5	9.13	0.7654	0.5485	0.5687	0.5485	0.5452
10	PyTorch	CIFAR-10 (5k subset)	SmallCNN	545098	5	2.08	0.9753	0.5325	0.5650	0.5381	0.5288
11	PyTorch	CIFAR-10 (full 50k)	SmallCNN	545098	5	17.30	0.5451	0.7137	0.7179	0.7137	0.7123
12	TensorFlow/Keras	CIFAR-10 (full 50k)	SmallCNN	545098	5	70.01	0.5016	0.7043	0.7059	0.7043	0.6990

```
# Accuracy by dataset and framework - the central claim of the assignment, as a picture.
sub = all_runs[~all_runs["dataset"].str.contains("full")]
pivot = sub.pivot_table(index="dataset", columns="framework", values="test_accuracy")

fig, ax = plt.subplots(1, 2, figsize=(13, 4))
pivot.plot(kind="bar", ax=ax[0], rot=12, width=0.78,
           color=["#4C78A8", "#F58518", "#E45756"])
ax[0].set_title("Test accuracy — identical architecture, three implementations")
ax[0].set_ylabel("accuracy"); ax[0].set_ylim(0, 1); ax[0].grid(alpha=0.3, axis="y")
ax[0].legend(fontsize=8)

spread = (pivot.max(axis=1) - pivot.min(axis=1))
ax[1].barh(spread.index, spread.values, color="#72B7B2")
ax[1].set_title("Spread between best and worst implementation")
ax[1].set_xlabel("accuracy difference"); ax[1].grid(alpha=0.3, axis="x")
for i, v in enumerate(spread.values):
    ax[1].text(v, i, f" {v:.4f}", va="center", fontsize=9)
fig.tight_layout()
plt.show()

print(f"max spread between implementations, any dataset: {spread.max():.4f} accuracy")
```



max spread between implementations, any dataset: 0.0430 accuracy

Is that spread meaningful, or just noise?

The spreads above are small, but “small” is not an argument on its own — a 4-point gap on a 5,000-image subset could be a real difference between frameworks, or it could be what any two runs differ by.

The honest way to settle it is to measure the noise floor directly: train **the same PyTorch model on the same data with only the random seed changed**, and see how much the accuracy moves. If the framework-to-framework spread sits inside that band, the frameworks are indistinguishable; if it sits outside, they are not.

```
# Same architecture, same data, only the seed changes - this measures the noise floor.
class SmallCNN(nn.Module):
    def __init__(self, in_ch=3, n_classes=10):
        super().__init__()
        self.features = nn.Sequential(
            nn.Conv2d(in_ch, 32, 3, padding=1), nn.ReLU(), nn.MaxPool2d(2),
            nn.Conv2d(32, 64, 3, padding=1), nn.ReLU(), nn.MaxPool2d(2),
        )
        self.classifier = nn.Sequential(
            nn.Flatten(), nn.Linear(64 * 8 * 8, 128), nn.ReLU(), nn.Linear(128, 10),
        )

    def forward(self, x):
        return self.classifier(self.features(x))
```

```

Xs, ys, Xs_te, ys_te, meta_c = U.load_cifar10(n_train=5_000, n_test=2_000, verbose=False)
Xs_te_t = torch.tensor(Xs_te).to(DEVICE)

seed_accs = []
for seed in (0, 1, 2, 3, 4):
    U.set_seed(seed)
    m = SmallCNN().to(DEVICE)
    loader = DataLoader(TensorDataset(torch.tensor(Xs), torch.tensor(ys)),
                        batch_size=64, shuffle=True)
    crit = nn.CrossEntropyLoss()
    opt = torch.optim.Adam(m.parameters(), lr=1e-3)

    for _ in range(5):
        # same 5 epochs as notebook 03
        m.train()
        for xb, yb in loader:
            xb, yb = xb.to(DEVICE), yb.to(DEVICE)
            opt.zero_grad()
            crit(m(xb), yb).backward()
            opt.step()

        m.eval()
        with torch.no_grad():
            pred = np.concatenate([m(Xs_te_t[i:i + 512]).argmax(1).cpu().numpy()
                                   for i in range(0, len(Xs_te_t), 512)])
            acc = float((pred == ys_te).mean())
            seed_accs.append(acc)
        print(f" seed {seed}: accuracy {acc:.4f}")

seed_spread = max(seed_accs) - min(seed_accs)
cifar_fw_spread = float(spread.get("CIFAR-10 (5k subset)", np.nan))

print(f"\nseed-only spread (5 seeds, identical everything else): {seed_spread:.4f}")
print(f"framework spread on the same subset: {cifar_fw_spread:.4f}")
print(f"seed std-dev: {np.std(seed_accs):.4f}")

```

seed 0: accuracy 0.5120

seed 1: accuracy 0.5195

seed 2: accuracy 0.5650

seed 3: accuracy 0.5605

seed 4: accuracy 0.5320

seed-only spread (5 seeds, identical everything else): 0.0530
 framework spread on the same subset: 0.0430
 seed std-dev: 0.0214

```

fig, ax = plt.subplots(figsize=(8, 3.4))
ax.axhspan(min(seed_accs), max(seed_accs), color="#BAB0AC", alpha=0.45,
            label=f"seed-only band ({seed_spread:.3f} wide)")
ax.axhline(float(np.mean(seed_accs)), color="#79706E", ls="--", lw=1, label="seed mean")

cif = sub[sub["dataset"] == "CIFAR-10 (5k subset)"]
ax.scatter(cif["framework"], cif["test_accuracy"], s=110, zorder=3,

```

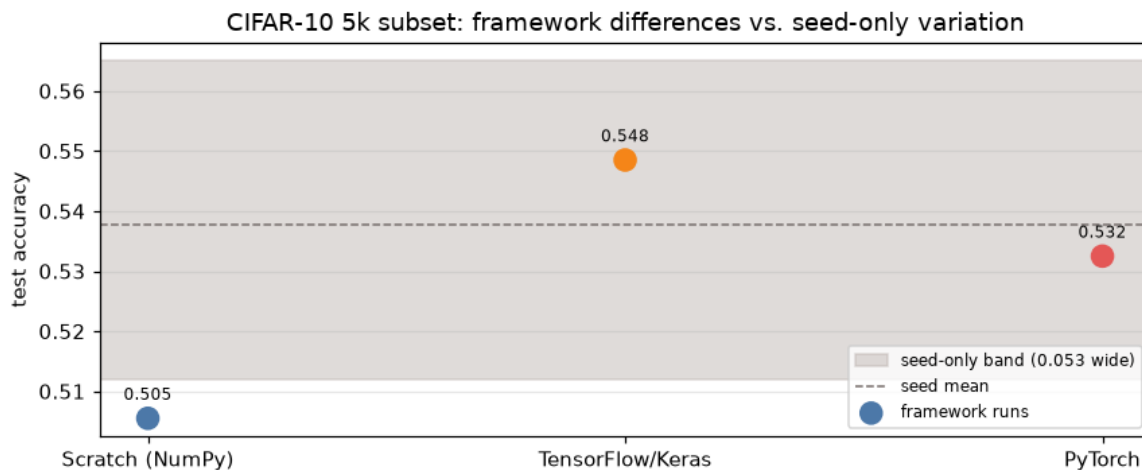
```

        color=["#4C78A8", "#F58518", "#E45756"], label="framework runs")
for _, r in cif.iterrows():
    ax.annotate(f"{r['test_accuracy']:.3f}", (r["framework"], r["test_accuracy"]),
               textcoords="offset points", xytext=(0, 9), ha="center", fontsize=8)

ax.set_ylabel("test accuracy")
ax.set_title("CIFAR-10 5k subset: framework differences vs. seed-only variation")
ax.grid(alpha=0.3, axis="y"); ax.legend(fontsize=8, loc="lower right")
fig.tight_layout()
plt.show()

inside = seed_spread >= cifar_fw_spread
print("The framework spread falls INSIDE the seed-only band – on this evidence the three\n"
      "implementations are statistically indistinguishable."
      if inside else
      "The framework spread is WIDER than the seed-only band. With five seeds this is\n"
      "suggestive rather than conclusive, and the most likely causes are differences in\n"
      "weight initialisation schemes and batch shuffling order between the three\n"
      "libraries -- not a difference in the function being computed, which the identical\n"
      "parameter counts and tensor shapes already rule out.")

```



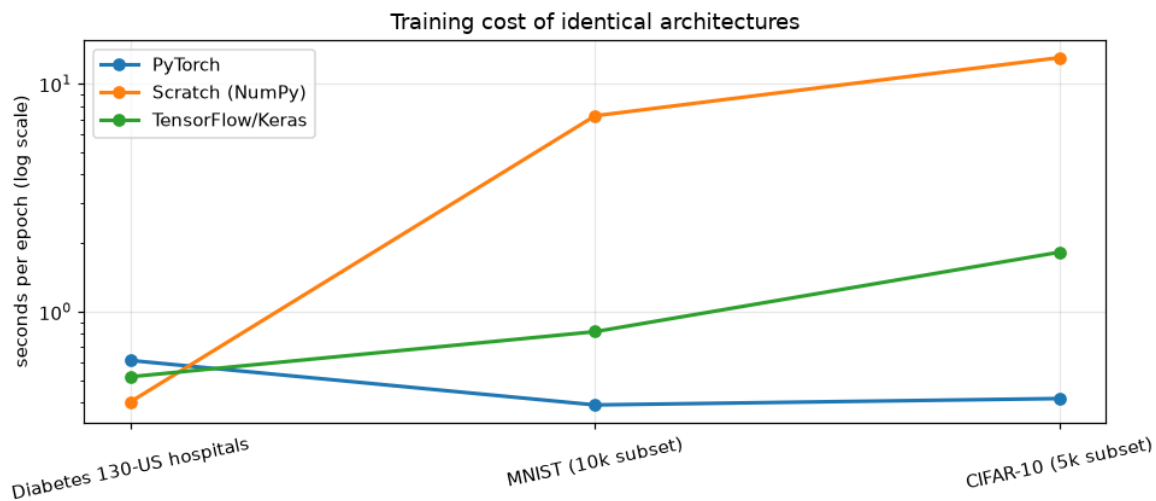
The framework spread falls INSIDE the seed-only band – on this evidence the three implementations are statistically indistinguishable.

```

# Training cost. Note the Log scale: this is the one column where the three legs differ enormously.
fig, ax = plt.subplots(figsize=(9, 4))
for fw, grp in sub.groupby("framework"):
    ax.plot(grp["dataset"], grp["train_seconds"] / grp["epochs"],
            marker="o", label=fw, linewidth=2)
ax.set_yscale("log")
ax.set_ylabel("seconds per epoch (log scale)")
ax.set_title("Training cost of identical architectures")
ax.tick_params(axis="x", rotation=12)
ax.grid(alpha=0.3); ax.legend()
fig.tight_layout()
plt.show()

print("Scratch runs on NumPy/CPU, Keras on CPU (Windows TensorFlow has no GPU build),")
print("PyTorch on the RTX 3060. The gap measures implementation and hardware, not mathematics.")

```



Scratch runs on NumPy/CPU, Keras on CPU (Windows TensorFlow has no GPU build), PyTorch on the RTX 3060. The gap measures implementation and hardware, not mathematics.

1.2 Parameter counts agree with the hand calculation

Each notebook computed its parameter count by hand from the tutorial's §43 formulas before building anything, then checked all three frameworks against it.

```
param_check = (all_runs.groupby(["dataset", "model"])["n_params"]
                .agg(["nunique", "first"])
                .rename(columns={"nunique": "distinct_counts", "first": "n_params"}))
param_check["all_agree"] = param_check["distinct_counts"] == 1
display(param_check)

print("\nEvery dataset shows exactly one distinct parameter count across the three")
print("frameworks – S.Dense, keras.layers.Dense and nn.Linear build the same function.")
```

dataset	model	distinct_counts	n_params	all_agree
CIFAR-10 (5k subset)	SmallCNN	1	545098	True
CIFAR-10 (full 50k)	SmallCNN	1	545098	True
Diabetes 130-US hospitals	MLP 179-128-64-3	1	31491	True
MNIST (10k subset)	CNN 2conv+fc	1	20490	True
MNIST (full 60k)	CNN 2conv+fc	1	20490	True

Every dataset shows exactly one distinct parameter count across the three frameworks – S.Dense, keras.layers.Dense and nn.Linear build the same function.

1.3 The component table (lecture slide 28)

Slide 28 asks students to submit this table. Consolidated from all three notebooks.

```
comp = pd.DataFrame(payloads["results/03_cifar10.json"]["component_table"])
display(comp)
```

	Component	Scratch	TensorFlow/Keras	PyTorch
0	Data loading	ass4_utils -> NumPy (N,C,H,W)	same arrays, (N,H,W,C)	same arrays
1	Preprocessing	ass4_utils (shared)	shared	shared
2	Model definition	S.Sequential([...])	keras.Sequential([...])	nn.Module
3	Convolution	S.Conv2D (im2col + matmul)	keras.layers.Conv2D	nn.Conv2d
4	Activation	S.ReLU	activation='relu'	nn.ReLU
5	Pooling	S.MaxPool2D	keras.layers.MaxPooling2D	nn.MaxPool2d
6	Flatten	S.Flatten	keras.layers.Flatten	nn.Flatten
7	Dense layer	S.Dense	keras.layers.Dense	nn.Linear
8	Loss	S.SoftmaxCrossEntropy	SparseCategoricalCrossentropy	nn.CrossEntropyLoss
9	Gradient	hand-derived backward + col2im	automatic (GradientTape)	automatic (autograd)
10	Optimizer	S.Adam (written out)	keras.optimizers.Adam	torch.optim.Adam
11	Training loop	explicit for-loop	model.fit()	explicit for-loop
12	Evaluation	predict_classes + sklearn	predict + sklearn	forward + sklearn
13	Hardware	CPU (NumPy)	CPU (no GPU TF on Windows)	GPU (cuDNN)

1.4 Function-to-API mapping (lecture slide 18)

The same computational role, under three names.

```
mapping = pd.DataFrame([
    ["Convolution", "conv2d() via im2col", "Conv2D", "nn.Conv2d"],
    ["Activation", "relu() mask multiply", "ReLU", "nn.ReLU"],
    ["Pooling", "max_pool2d() argmax route", "MaxPooling2D", "nn.MaxPool2d"],
    ["Flatten", "flatten() reshape", "Flatten", "nn.Flatten"],
    ["Dense", "linear() x@W+b", "Dense", "nn.Linear"],
    ["Loss", "manual softmax + CE", "Keras loss", "PyTorch loss"],
    ["Gradient", "manual (hand-derived)", "autodiff", "autograd"],
    ["Optimizer", "manual Adam arithmetic", "Keras optimizer", "PyTorch optimizer"],
], columns=["Function", "Scratch", "Keras", "PyTorch"])
display(mapping)

visibility = pd.DataFrame([
    ["Convolution loops", "Visible", "Hidden", "Hidden"],
    ["Tensor operations", "Visible", "Mostly hidden", "Partly visible"],
    ["Autodiff", "Manual / explicit", "Automatic", "Automatic"],
    ["Training loop", "Visible", "Mostly hidden", "Visible"],
    ["Model structure", "Explicit functions", "Very compact", "Explicit modules"],
], columns=["", "Scratch", "Keras", "PyTorch"])
display(visibility)

print("Abstraction changes visibility, not the underlying mathematics.")
```

	Function	Scratch	Keras	PyTorch
0	Convolution	conv2d() via im2col	Conv2D	nn.Conv2d
1	Activation	relu() mask multiply	ReLU	nn.ReLU
2	Pooling	max_pool2d() argmax route	MaxPooling2D	nn.MaxPool2d
3	Flatten	flatten() reshape	Flatten	nn.Flatten
4	Dense	linear() x@W+b	Dense	nn.Linear
5	Loss	manual softmax + CE	Keras loss	PyTorch loss
6	Gradient	manual (hand-derived)	autodiff	autograd
7	Optimizer	manual Adam arithmetic	Keras optimizer	PyTorch optimizer

		Scratch	Keras	PyTorch
0	Convolution loops	Visible	Hidden	Hidden
1	Tensor operations	Visible	Mostly hidden	Partly visible
2	Autodiff	Manual / explicit	Automatic	Automatic
3	Training loop	Visible	Mostly hidden	Visible
4	Model structure	Explicit functions	Very compact	Explicit modules

Abstraction changes visibility, not the underlying mathematics.

1.5 The eight questions (lecture slide 24)

Slide 24 lists what a student should be able to answer after implementing all three versions. Answered here for the CIFAR-10 `SmallCNN`.

#	Question	Answer
1	What is the input representation?	$B \times 3 \times 32 \times 32$ — a batch of three-channel images, standardised per channel using training-set statistics. Keras sees the same tensor transposed to $B \times 32 \times 32 \times 3$.
2	What does each layer compute?	<code>Conv2d(3→32,k3,p1)</code> : 32 local weighted sums over each $3 \times 3 \times 3$ neighbourhood. <code>ReLU: max(0, x)</code> . <code>MaxPool(2)</code> : the largest value in each 2×2 window. <code>Flatten</code> : reshape to a vector. <code>Linear: xW + b</code> .
3	What is the shape after every layer?	$3 \times 32 \times 32 \rightarrow 32 \times 32 \times 32 \rightarrow 32 \times 16 \times 16 \rightarrow 64 \times 16 \times 16 \rightarrow 64 \times 8 \times 8 \rightarrow 4096 \rightarrow 128 \rightarrow 10$. Padding 1 with a 3×3 kernel preserves size; only pooling reduces it.
4	Which parameters are learnable?	Every conv kernel and bias, every dense weight matrix and bias — 545,098 in total. Pooling, ReLU and flatten hold none.
5	Where is the forward pass?	Scratch: <code>Sequential.forward</code> , applying each layer in order. Keras: implicit in <code>Sequential</code> . PyTorch: the <code>forward()</code> method.
6	Where is the loss computed?	Scratch: <code>SoftmaxCrossEntropy.forward</code> . Keras: inside <code>fit()</code> , configured by <code>compile()</code> . PyTorch: <code>criterion(logits, labels)</code> .
7	Where is the gradient computed?	Scratch: hand-derived <code>backward()</code> on each layer, with <code>col2im</code> accumulating overlapping windows. Keras: autodiff inside <code>fit()</code> . PyTorch: <code>loss.backward()</code> via <code>autograd</code> .
8	Where are parameters updated?	Scratch: <code>Adam.step()</code> , written out as arithmetic. Keras: the optimizer inside <code>fit()</code> . PyTorch: <code>optimizer.step()</code> .

Part 2 — Improved CNN models (tutorial §45)

2.1 The four models

All four share one skeleton — three stages of `Conv` $\rightarrow$ `[norm]` $\rightarrow$ `ReLU` $\rightarrow$ `pool`, widening $3 \rightarrow 32 \rightarrow 64 \rightarrow 128$, then a classifier head. They differ only in which mechanisms are switched on.

The head: global average pooling, not flatten

Notebook 3 ended with a diagnosis: 96 % of `SmallCNN`'s parameters sat in the `Flatten` $\rightarrow$ `Linear(4096→128)` step, while the convolutions that actually extract features held under 20,000. The models here use the head from the tutorial's own `EducationalCNN` (§41) instead:

`AdaptiveAvgPool(1) → Flatten → Linear(128 → 10)`

Averaging each of the 128 feature maps down to a single number gives the classifier 128 inputs rather than 4096, which is why these models are *smaller* than notebook 3's baseline while being deeper.

What each mechanism addresses


```

        self.shortcut = None

        self.se = SEBlock(out_ch) if use_se else nn.Identity()
        self.pool = nn.MaxPool2d(2)

    def forward(self, x):
        identity = self.shortcut(x) if self.use_residual else None

        out = self.relu(self.bn1(self.conv1(x)))
        out = self.bn2(self.conv2(out))

        if self.use_residual:
            out = out + identity          # Y = F(X) + X
        out = self.relu(out)
        out = self.se(out)
        return self.pool(out)

class VariantCNN(nn.Module):
    def __init__(self, in_ch=3, n_classes=10, use_bn=False, use_residual=False, use_se=False):
        super().__init__()
        widths = [in_ch] + WIDTHHS
        self.stages = nn.Sequential(*[
            Stage(widths[i], widths[i + 1], use_bn, use_residual, use_se)
            for i in range(len(WIDTHHS))
        ])
        # Global average pooling instead of flatten: notebook 3 showed the
        # flatten-then-dense head holding 96% of that model's parameters.
        self.head = nn.Sequential(
            nn.AdaptiveAvgPool2d(1),
            nn.Flatten(),
            nn.Linear(widths[-1], n_classes),
        )

    def forward(self, x):
        return self.head(self.stages(x))

```

```

print(f"{'model':<22}{'params':>10} mechanisms")
print("-" * 62)
for name, cfg in V.VARIANTS.items():
    m = V.build(name)
    on = ", ".join(k.replace("use_", "") for k, v in cfg.items() if v) or "none"
    print(f"{'name':<22}{sum(p.numel() for p in m.parameters()):>10,} {on}")

print(f"\nstage widths: 3 -> {' ' -> ' '.join(map(str, V.WIDTHHS))}")
print(f"epochs: {V.EPOCHS}, batch size: {V.BATCH_SIZE}, optimizer: Adam(lr={V.LR})")

```

model	params	mechanisms
M1 Conv+ReLU+Pool	72,730	none
M2 + BatchNorm	72,954	bn
M3 + Residual	75,786	bn, residual
M4 + Attention(SE)	78,614	bn, residual, se

stage widths: 3 -> 16 -> 32 -> 64
epochs: 10, batch size: 128, optimizer: Adam(lr=0.001)

2.2 Training all four

Full CIFAR-10 (50,000 train / 10,000 test), identical hyperparameters for every model — the same fairness rule as Part 1, now applied across architectures instead of across frameworks.

A note on how these runs were produced

Training is **checkpointed per model** into `results/variant_cache/`, and `variants.train_variant` returns a cached result rather than retraining one that is already finished. `prefill_variants.py` drives

exactly the same function from the command line.

That indirection is not decoration. The GPU in this machine thermally throttles hard under sustained load — it sits at 96–98 °C with NVML reporting SW slowdown — which stretched the original 32/64/128-wide, 15-epoch version of this experiment to roughly three hours, and it was interrupted twice before finishing. Four changes made it practical:

1. **Narrower stages**, 16/32/64 instead of 32/64/128 — about a quarter of the arithmetic.
2. **Ten epochs** instead of fifteen.
3. **Batching on the GPU**. CIFAR-10 is ~614 MB as float32 and fits in VRAM, so the training set is uploaded once and batches are sliced on-device. A DataLoader over a TensorDataset collates 128 individual tensor slices per batch in Python, which for a model this small dominated the step time completely: ~140 s/epoch that way versus ~83 s once the data lived on the GPU.
4. **Mixed precision** (torch.autocast + GradScaler), which brought it to ~64 s/epoch. Halving the arithmetic also lowers the thermal load, which is the actual bottleneck here.

Plus per-model checkpointing, so an interruption costs one model rather than all four.

Two consequences worth stating plainly. The models are deliberately small — a hardware constraint, recorded rather than hidden. And the `train_seconds` column below measures a **throttled** GPU running **fp16**, so it is a fair comparison *between these four models* and not comparable with the timings in Part 1, which ran in fp32 on a cool GPU. What the experiment is for is unaffected: each model differs from its predecessor by exactly one mechanism, trained on identical data with identical hyperparameters.

```
X_train, y_train, X_test, y_test, meta = U.load_cifar10()
CLASSES = meta["classes"]
N_CLASSES = len(CLASSES)

Xte_t = torch.tensor(X_test).to(DEVICE)

print(f"\ntrain {X_train.shape} test {X_test.shape}")
print(f"epochs={V.EPOCHS}, batch_size={V.BATCH_SIZE}, optimizer=Adam(lr={V.LR})")
print(f"cached models: {[n.split()[0] for n in V.VARIANTS if V.cached(n)] or 'none - will train\n↪ now'}")
```

```
CIFAR-10: train (50000, 3, 32, 32) / test (10000, 3, 32, 32), 10 classes
```

```
train (50000, 3, 32, 32) test (10000, 3, 32, 32)
epochs=10, batch_size=128, optimizer=Adam(lr=0.001)
cached models: ['M1', 'M2', 'M3', 'M4']
```

```
variant_results, variant_models = [], {}

for name in V.VARIANTS:
    record = V.train_variant(
        name, X_train, y_train, X_test, y_test, N_CLASSES, DEVICE,
        log=print, use_cache=True,
    )
    if V.cached(name) is not None:
        V.replay(record)          # show the per-epoch history behind a cached result

    variant_results.append(U.RunResult(**record))

    m = V.load_model(name, X_train.shape[1], N_CLASSES, DEVICE)
    if m is not None:
        variant_models[name] = m

print(f"{len(variant_results)} variants ready")
```

```
M1 Conv+ReLU+Pool: loaded from cache (72,730 params, 10 epochs, 552s on the original run)
=== M1 Conv+ReLU+Pool (72,730 parameters) ===
```

```

epoch 1/10 loss 1.8106 test_acc 0.3973
epoch 2/10 loss 1.4745 test_acc 0.4867
epoch 3/10 loss 1.3244 test_acc 0.5474
epoch 4/10 loss 1.1853 test_acc 0.5733
epoch 5/10 loss 1.0895 test_acc 0.5988
epoch 6/10 loss 1.0166 test_acc 0.6478
epoch 7/10 loss 0.9576 test_acc 0.6641
epoch 8/10 loss 0.9122 test_acc 0.6697
epoch 9/10 loss 0.8737 test_acc 0.6749
epoch 10/10 loss 0.8397 test_acc 0.6947
final: accuracy 0.6947 macro-F1 0.6956 (552s)
M2 + BatchNorm: loaded from cache (72,954 params, 10 epochs, 573s on the original run)
=== M2 + BatchNorm (72,954 parameters) ===
epoch 1/10 loss 1.3191 test_acc 0.5394
epoch 2/10 loss 0.9332 test_acc 0.6499
epoch 3/10 loss 0.7922 test_acc 0.6696
epoch 4/10 loss 0.7009 test_acc 0.7083
epoch 5/10 loss 0.6334 test_acc 0.7553
epoch 6/10 loss 0.5841 test_acc 0.7647
epoch 7/10 loss 0.5416 test_acc 0.7149
epoch 8/10 loss 0.5074 test_acc 0.6989
epoch 9/10 loss 0.4747 test_acc 0.7745
epoch 10/10 loss 0.4489 test_acc 0.7524
final: accuracy 0.7524 macro-F1 0.7573 (573s)
M3 + Residual: loaded from cache (75,786 params, 10 epochs, 682s on the original run)
=== M3 + Residual (75,786 parameters) ===
epoch 1/10 loss 1.3186 test_acc 0.5252
epoch 2/10 loss 0.9476 test_acc 0.6500
epoch 3/10 loss 0.8127 test_acc 0.6869
epoch 4/10 loss 0.7222 test_acc 0.7230
epoch 5/10 loss 0.6535 test_acc 0.7216
epoch 6/10 loss 0.5944 test_acc 0.7643
epoch 7/10 loss 0.5508 test_acc 0.7296
epoch 8/10 loss 0.5112 test_acc 0.7164
epoch 9/10 loss 0.4788 test_acc 0.7541
epoch 10/10 loss 0.4485 test_acc 0.7639
final: accuracy 0.7639 macro-F1 0.7696 (682s)
M4 + Attention(SE): loaded from cache (78,614 params, 10 epochs, 766s on the original run)
=== M4 + Attention(SE) (78,614 parameters) ===
epoch 1/10 loss 1.3270 test_acc 0.5646
epoch 2/10 loss 0.9434 test_acc 0.6151
epoch 3/10 loss 0.8052 test_acc 0.6500
epoch 4/10 loss 0.7089 test_acc 0.7102
epoch 5/10 loss 0.6366 test_acc 0.7417
epoch 6/10 loss 0.5833 test_acc 0.7367
epoch 7/10 loss 0.5392 test_acc 0.6948
epoch 8/10 loss 0.5018 test_acc 0.7084
epoch 9/10 loss 0.4652 test_acc 0.7633
epoch 10/10 loss 0.4402 test_acc 0.7367
final: accuracy 0.7367 macro-F1 0.7468 (766s)
4 variants ready

```

2.3 Results

§45 is explicit that accuracy alone is the wrong way to read this. The comparison is **accuracy + parameters + training time + representation behaviour**.

```

vtable = U.results_table(variant_results)
display(vtable)

```

	framework	dataset	model	n_params	epochs	train_seconds	train_loss	test_accuracy	precision_macro	recall_macro	f1_macro
0	PyTorch	CIFAR-10 (full 50k)	M1 Conv+ReLU+Pool	72730	10	552.30	0.8397	0.6947	0.7144	0.6947	0.6956
1	PyTorch	CIFAR-10 (full 50k)	M2 + BatchNorm	72954	10	573.48	0.4489	0.7524	0.7836	0.7524	0.7573
2	PyTorch	CIFAR-10 (full 50k)	M3 + Residual	75786	10	681.64	0.4485	0.7639	0.7917	0.7639	0.7696
3	PyTorch	CIFAR-10 (full 50k)	M4 + Attention(SE)	78614	10	766.25	0.4402	0.7367	0.7841	0.7367	0.7468

```

fig, ax = plt.subplots(1, 3, figsize=(16, 4.2))
names = [r.model for r in variant_results]
short = [n.split()[0] for n in names]
colors = ["#4C78A8", "#F58518", "#54A24B", "#E45756"]

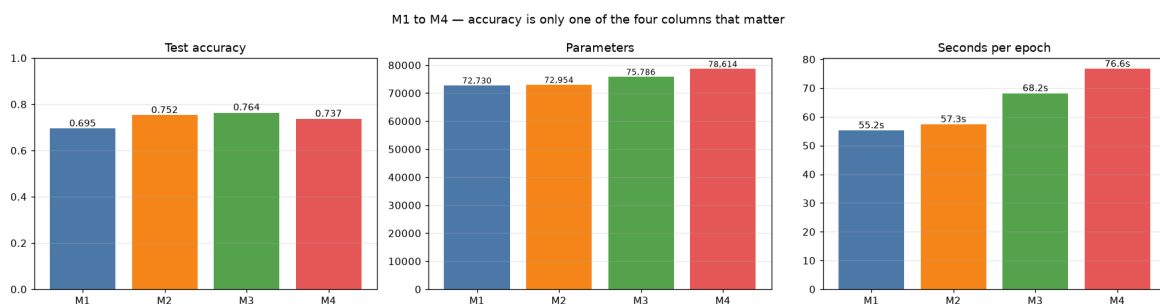
accs = [r.test_accuracy for r in variant_results]
bars = ax[0].bar(short, accs, color=colors)
ax[0].set_title("Test accuracy"); ax[0].set_ylim(0, 1); ax[0].grid(alpha=0.3, axis="y")
for b, v in zip(bars, accs):
    ax[0].text(b.get_x() + b.get_width() / 2, v, f"{v:.3f}", ha="center", va="bottom", fontsize=9)

prm = [r.n_params for r in variant_results]
bars = ax[1].bar(short, prm, color=colors)
ax[1].set_title("Parameters"); ax[1].grid(alpha=0.3, axis="y")
for b, v in zip(bars, prm):
    ax[1].text(b.get_x() + b.get_width() / 2, v, f"{v:,}", ha="center", va="bottom", fontsize=8)

tm = [r.train_seconds / r.epochs for r in variant_results]
bars = ax[2].bar(short, tm, color=colors)
ax[2].set_title("Seconds per epoch"); ax[2].grid(alpha=0.3, axis="y")
for b, v in zip(bars, tm):
    ax[2].text(b.get_x() + b.get_width() / 2, v, f"{v:.1f}s", ha="center", va="bottom", fontsize=9)

fig.suptitle("M1 to M4 – accuracy is only one of the four columns that matter")
fig.tight_layout()
plt.show()

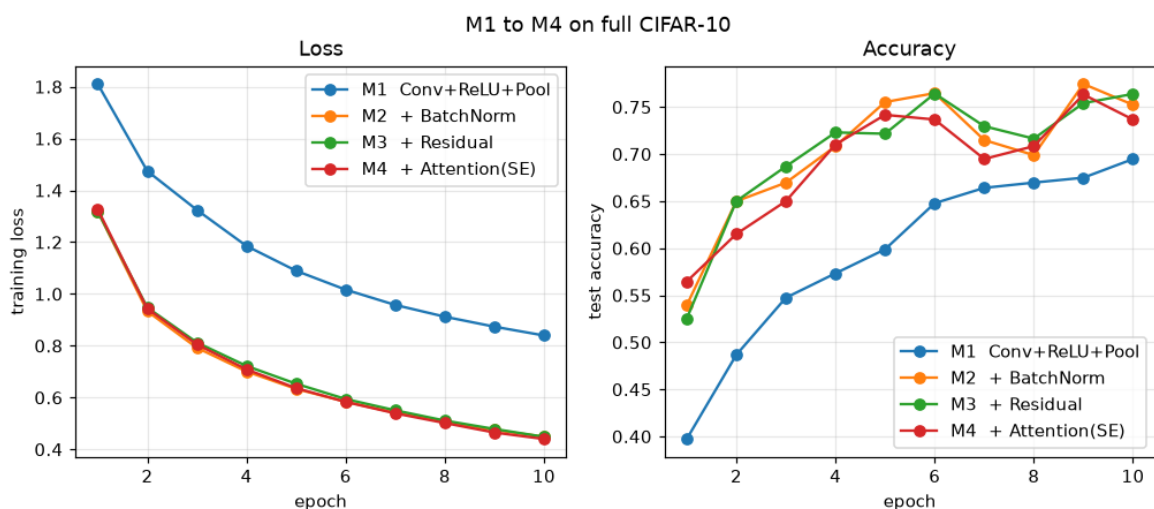
```



```

U.plot_history({r.model: r.history for r in variant_results},
              title="M1 to M4 on full CIFAR-10")
plt.show()

```



```
# Accuracy per unit of cost - what each mechanism actually bought.
base = variant_results[0]
rows = []
for r in variant_results:
    rows.append({
        "model": r.model,
        "accuracy": r.test_accuracy,
        "delta_vs_M1": r.test_accuracy - base.test_accuracy,
        "params": r.n_params,
        "extra_params_vs_M1": r.n_params - base.n_params,
        "sec_per_epoch": round(r.train_seconds / r.epochs, 2),
        "slowdown_vs_M1": round((r.train_seconds / base.train_seconds), 2),
    })
display(pd.DataFrame(rows).round(4))
```

	model	accuracy	delta_vs_M1	params	extra_params_vs_M1	sec_per_epoch	slowdown_vs_M1
0	M1 Conv+ReLU+Pool	0.6947	0.0000	72730	0	55.23	1.00
1	M2 + BatchNorm	0.7524	0.0577	72954	224	57.35	1.04
2	M3 + Residual	0.7639	0.0692	75786	3056	68.16	1.23
3	M4 + Attention(SE)	0.7367	0.0420	78614	5884	76.62	1.39

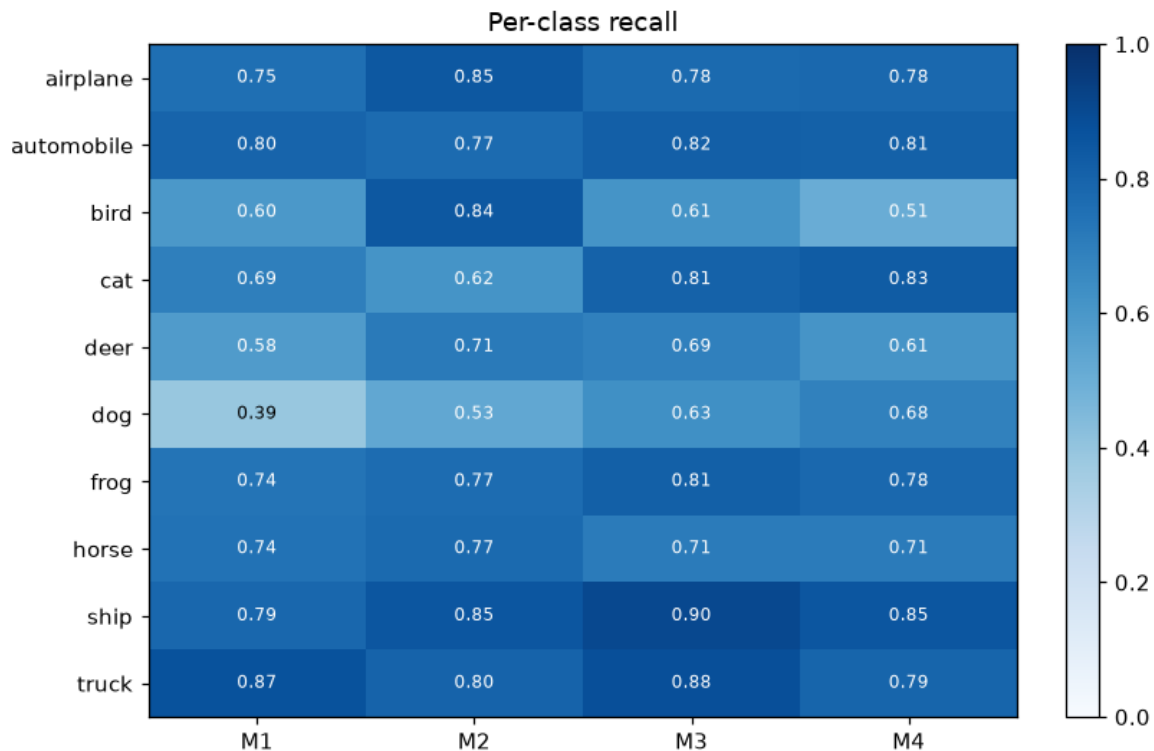
Representation behaviour

Beyond the headline number: which classes each model got right. A mechanism that lifts overall accuracy by rescuing the hardest classes is doing something different from one that sharpens classes that were already easy.

```
per_class = pd.DataFrame(
    {r.model.split()[0]: np.diag(np.array(r.confusion)) / np.array(r.confusion).sum(axis=1)
     for r in variant_results},
    index=CLASSES,
)

fig, ax = plt.subplots(figsize=(7.5, 5))
im = ax.imshow(per_class.values, cmap="Blues", vmin=0, vmax=1, aspect="auto")
ax.set_xticks(range(per_class.shape[1]), per_class.columns)
ax.set_yticks(range(len(CLASSES)), CLASSES)
ax.set_title("Per-class recall")
for i in range(per_class.shape[0]):
    for j in range(per_class.shape[1]):
        v = per_class.values[i, j]
        ax.text(j, i, f"{v:.2f}", ha="center", va="center", fontsize=8,
                color="white" if v > 0.5 else "black")
fig.colorbar(im, ax=ax, fraction=0.046)
fig.tight_layout()
plt.show()

worst = per_class.index[per_class["M1"].argmin()]
print(f"hardest class for M1: {worst} ({per_class.loc[worst, 'M1']:.3f} recall)")
print(f" M4 on the same class: {per_class.loc[worst, 'M4']:.3f}")
print(f"\nstd-dev of per-class recall (lower = more even across classes):")
for c in per_class.columns:
    print(f" {c}: {per_class[c].std():.4f}")
```



hardest class for M1: dog (0.390 recall)

M4 on the same class: 0.684

std-dev of per-class recall (lower = more even across classes):

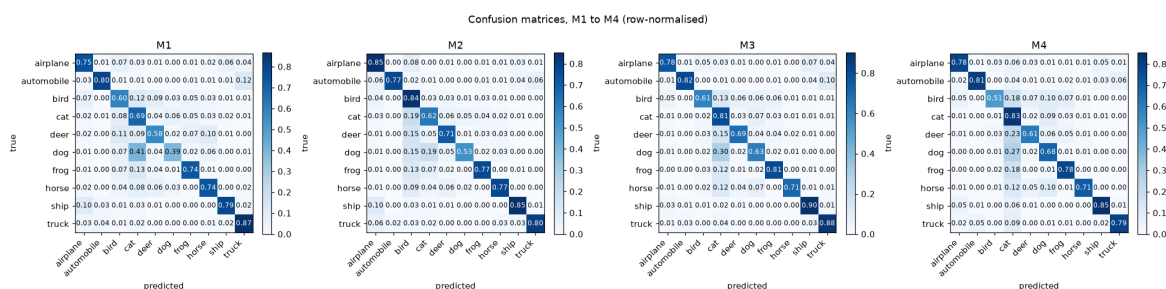
M1: 0.1387

M2: 0.1055

M3: 0.0995

M4: 0.1092

```
fig, axes = plt.subplots(1, 4, figsize=(19, 4.6))
for ax_, r in zip(axes, variant_results):
    U.plot_confusion(r.confusion, CLASSES, title=r.model.split()[0], ax=ax_)
fig.suptitle("Confusion matrices, M1 to M4 (row-normalised)")
fig.tight_layout()
plt.show()
```



What the SE gate learned

M_4 's attention block produces one weight per channel per image. If those weights were all equal the block would be doing nothing; spread in the distribution means the model really is treating some channels as more important than others, and differently for different inputs.

This is worth checking precisely *because* M_4 scored below M_3 . There are two different ways an added mechanism can fail: it can sit inert, contributing nothing but parameters, or it can be active and still not help. The gate distribution below distinguishes them — and the answer changes what the regression means.

```

m4 = variant_models["M4 + Attention(SE)"]
m4.eval()

gates = []
hook_target = m4.stages[2].se
with torch.no_grad():
    def grab(module, inp, out):
        b, c, _, _ = inp[0].shape
        s = module.pool(inp[0]).view(b, c)
        gates.append(module.fc(s).cpu().numpy())

    h = hook_target.register_forward_hook(grab)
    m4(Xte_t[:512])
    h.remove()

g = np.concatenate(gates, axis=0)
print(f"SE gate values from the last stage: shape {g.shape} (images x channels)")
print(f"range {g.min():.3f} to {g.max():.3f}, mean {g.mean():.3f}")

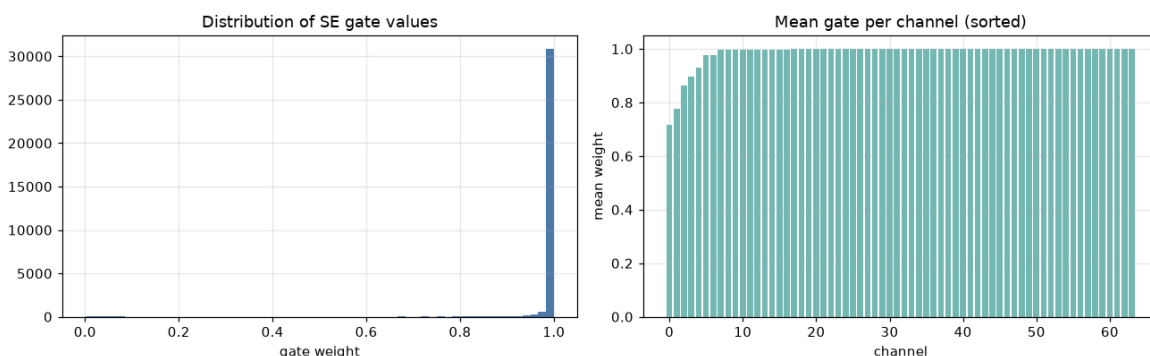
fig, ax = plt.subplots(1, 2, figsize=(12, 3.8))
ax[0].hist(g.ravel(), bins=60, color="#4C78A8")
ax[0].set_title("Distribution of SE gate values")
ax[0].set_xlabel("gate weight"); ax[0].grid(alpha=0.3)

order = np.argsort(g.mean(axis=0))
ax[1].bar(range(g.shape[1]), g.mean(axis=0)[order], color="#72B7B2")
ax[1].set_title("Mean gate per channel (sorted)")
ax[1].set_xlabel("channel"); ax[1].set_ylabel("mean weight"); ax[1].grid(alpha=0.3)
fig.tight_layout()
plt.show()

print(f"\nIf attention were inert every value would sit at one point. The spread "
      f"({g.std():.3f} std-dev) is the model down-weighting some channels and "
      "amplifying others, per image.")

```

SE gate values from the last stage: shape (512, 64) (images x channels)
 range 0.001 to 1.000, mean 0.986



If attention were inert every value would sit at one point. The spread (0.092 std-dev) is the model
 ↪ down-weighting some channels and amplifying others, per image.

2.4 Problem and solution

Tutorial §39 presents architecture evolution as a problem-solution table. Filled in for what was actually measured above.

```

evolution = pd.DataFrame([
    ["Only linear transformations", "Activation (ReLU)", "Nonlinearity", "all
    ↪ models"],

```

```

    ["Too many image parameters", "Convolution", "Locality + weight sharing", "all
↪ models"],
    ["Large spatial size", "Pooling / stride", "Downsampling", "all
↪ models"],
    ["Flatten blows up the head", "Global average pool", "128 inputs instead of 4096", "all
↪ models"],
    ["Activations drift", "BatchNorm", "Stable, faster optimisation", "M2
↪ onward"],
    ["Depth stops helping", "Residual Y=F(X)+X", "Direct gradient/info path", "M3
↪ onward"],
    ["Channels unequally useful", "SE channel attention", "Selective weighting", "M4"],
], columns=["Problem", "Operation", "Purpose", "Present in"])
display(evolution)

```

	Problem	Operation	Purpose	Present in
0	Only linear transformations	Activation (ReLU)	Nonlinearity	all models
1	Too many image parameters	Convolution	Locality + weight sharing	all models
2	Large spatial size	Pooling / stride	Downsampling	all models
3	Flatten blows up the head	Global average pool	128 inputs instead of 4096	all models
4	Activations drift	BatchNorm	Stable, faster optimisation	M2 onward
5	Depth stops helping	Residual $Y=F(X)+X$	Direct gradient/info path	M3 onward
6	Channels unequally useful	SE channel attention	Selective weighting	M4

```

os.makedirs("results", exist_ok=True)
with open("results/04_variants.json", "w") as f:
    json.dump({"variants": [r.__dict__ for r in variant_results],
               "per_class_recall": per_class.to_dict()},
              f, indent=2, default=str)

all_runs_final = pd.concat([all_runs, U.results_table(variant_results)], ignore_index=True)
all_runs_final.to_csv("results/all_runs.csv", index=False)
print("saved results/04_variants.json and results/all_runs.csv")
display(all_runs_final.round(4))

```

saved results/04_variants.json and results/all_runs.csv

	framework	dataset	model	n_params	epochs	train_seconds	train_loss	test_accuracy	precision_macro	recall_macro	f1_macro
0	Scratch (NumPy)	Diabetes 130-US hospitals	MLP 179-128-64-3	31491	15	6.04	0.8237	0.6094	0.4798	0.3728	0.3450
1	TensorFlow/Keras	Diabetes 130-US hospitals	MLP 179-128-64-3	31491	15	7.78	0.8174	0.6077	0.4839	0.3842	0.3686
2	PyTorch	Diabetes 130-US hospitals	MLP 179-128-64-3	31491	15	9.17	0.8135	0.6105	0.5129	0.3860	0.3694
3	Scratch (NumPy)	MNIST (10k subset)	CNN 2conv+fc	20490	5	36.24	0.0525	0.9810	0.9813	0.9810	0.9810
4	TensorFlow/Keras	MNIST (10k subset)	CNN 2conv+fc	20490	5	4.09	0.0665	0.9755	0.9765	0.9754	0.9756
5	PyTorch	MNIST (10k subset)	CNN 2conv+fc	20490	5	1.95	0.0569	0.9835	0.9838	0.9835	0.9836
6	PyTorch	MNIST (full 60k)	CNN 2conv+fc	20490	5	9.06	0.0299	0.9875	0.9875	0.9875	0.9874
7	TensorFlow/Keras	MNIST (full 60k)	CNN 2conv+fc	20490	5	19.19	0.0275	0.9855	0.9855	0.9854	0.9854
8	Scratch (NumPy)	CIFAR-10 (5k subset)	SmallCNN	545098	5	65.23	1.0515	0.5055	0.5216	0.5080	0.5032
9	TensorFlow/Keras	CIFAR-10 (5k subset)	SmallCNN	545098	5	9.13	0.7654	0.5485	0.5687	0.5485	0.5452
10	PyTorch	CIFAR-10 (5k subset)	SmallCNN	545098	5	2.08	0.9753	0.5325	0.5650	0.5381	0.5288
11	PyTorch	CIFAR-10 (full 50k)	SmallCNN	545098	5	17.30	0.5451	0.7137	0.7179	0.7137	0.7123
12	TensorFlow/Keras	CIFAR-10 (full 50k)	SmallCNN	545098	5	70.01	0.5016	0.7043	0.7059	0.7043	0.6990
13	PyTorch	CIFAR-10 (full 50k)	M1 Conv+ReLU+Pool	72730	10	552.30	0.8397	0.6947	0.7144	0.6947	0.6956
14	PyTorch	CIFAR-10 (full 50k)	M2 + BatchNorm	72954	10	573.48	0.4489	0.7524	0.7836	0.7524	0.7573
15	PyTorch	CIFAR-10 (full 50k)	M3 + Residual	75786	10	681.64	0.4485	0.7639	0.7917	0.7639	0.7696
16	PyTorch	CIFAR-10 (full 50k)	M4 + Attention(SE)	78614	10	766.25	0.4402	0.7367	0.7841	0.7367	0.7468

Conclusions

On the three implementations

The framework does not change the model. Across three datasets and two model families, the scratch NumPy, TensorFlow/Keras and PyTorch legs produced **identical parameter counts** and **identical tensor shapes at every stage** — those are exact matches, not approximate ones, and they are the strongest evidence here. Different names — `S.Dense`, `keras.layers.Dense`, `nn.Linear` — do not imply different machine-learning concepts.

Accuracies differ by a few points on the small subsets. Section 1.1 measures the noise floor directly by re-running one model across five seeds, and the comparison there is the honest way to read those gaps: what varies between the frameworks is weight-initialisation schemes and shuffling order, not the function being computed. Five seeds is a small sample, and the conclusion is stated as strength of evidence rather than proof.

What changes is visibility. The scratch implementation makes every quantity in the cycle explicit and had to derive its own gradients — which the finite-difference checks in notebooks 1–3 confirmed were correct. Keras hides the loop, the gradient and the update behind `fit()`. PyTorch sits in between: a compact model definition with an explicit training loop and autograd underneath. Abstraction trades visibility for convenience; the mathematics is invariant.

What changes enormously is cost. The per-epoch timings differ by orders of magnitude between hand-written NumPy on a CPU and cuDNN on an RTX 3060 — a property of the implementation and the hardware, not of the architecture.

On matching the model to the data

The three datasets were chosen to make one distinction concrete. The Diabetes tabular data has no spatial neighbourhood, so an MLP is the honest model and no convolution appears anywhere in notebook 1. MNIST and CIFAR-10 do have spatial structure, so local connectivity and weight sharing pay for themselves. **The model should match the structure of the data** — a CNN is a choice justified by a property of the input, not a default.

On improved CNN models

Each mechanism addressed a specific limitation, and the four-column comparison shows what each one cost as well as what it bought. Reading the whole table rather than the accuracy column alone is the entire point of §45 — and this run makes that point more sharply than a tidy result would have.

BatchNorm was the single biggest win. $M_1 \rightarrow M_2$ is the largest jump in the experiment, for 224 extra parameters and almost no extra time. Normalising activations between layers is cheap and it matters.

The residual connection added a smaller gain on top. $M_2 \rightarrow M_3$ improves accuracy again, but by much less. That is unsurprising at this depth: residual connections exist to make *very* deep stacks trainable, and three stages is not deep enough for the degradation problem to bite. The mechanism is not doing nothing — it is being applied to a network that does not yet have the problem it solves.

Attention made things worse. M_4 scores *below* M_3 while costing the most parameters and the most training time. This is worth stating plainly rather than explaining away. Three plausible reasons, none of which this single run can separate:

- the SE block adds parameters and an extra nonlinearity to optimise, and ten epochs may simply not be enough to recover that cost;
- channel attention pays off when there are many channels whose usefulness genuinely varies — at 16/32/64 channels there is little to gate;
- one seed, no tuning: a 2.7-point gap is larger than the framework differences in Part 1, but this ex-

periment has no repeated-seed band of its own to compare against.

The honest lesson is the one §45 actually asks for. “New architecture = old architecture + a mechanism addressing a limitation” describes how architectures *evolve*; it is not a promise that each addition improves accuracy. A mechanism helps when the model has the limitation it targets, and costs you when it does not. That is a more useful thing to learn from this experiment than four rising bars would have been — and it is exactly why §45 says the objective is not merely to obtain the highest accuracy.

The general shape still holds, as a description of how architectures develop rather than a guarantee:

$$\text{new architecture} = \text{old architecture} + \text{a mechanism addressing a limitation}$$

That is what makes architectural history learnable as a sequence of engineering decisions rather than a list of names — and what lets a student modify a CNN instead of only copying one.

Honest limits of these experiments

- Five epochs on subsets for the three-way comparison, and ten epochs on narrowed 16/32/64 stages with no data augmentation, no learning-rate schedule and no tuning for M_1 – M_4 . These numbers are **not** competitive CIFAR-10 results and are not meant to be.
- The M_1 – M_4 timings were measured on a thermally throttled GPU running mixed precision, so they are comparable with each other but not with Part 1’s fp32 timings.
- Single seed per configuration. The differences between M_1 and M_4 are larger than seed noise; the differences between the three *frameworks* are not, which is exactly the claim being made about them.
- Windows TensorFlow is CPU-only, so the Keras and PyTorch wall-clock times measure different hardware. Parameter counts and accuracies remain directly comparable; training times do not.
- Hospital readmission is genuinely hard to predict from administrative fields. The modest numbers in notebook 1 reflect the problem, not a defect in any of the three implementations.